

Problems Session 7

Representation of Graphs

Problem 1

Give an adjacency-list representation for a complete binary tree on 7 vertices. Give an equivalent adjacency-matrix representation. Assume that vertices are numbered from 1 to 7 as in a binary heap.

Problem 2

The **transpose** of a directed graph $G = (V, E)$ is the graph $G^T = (V, E^T)$, where

$$E^T = \{(v, u) \in V \times V : (u, v) \in E\}.$$

Thus, G^T is G with all its edges reversed. Describe efficient algorithms for computing G^T from G , for both the adjacency-list and adjacency-matrix representations of G . Analyze the running times of your algorithms.

Problem 3

The *square* of a directed graph $G = (V, E)$ is the graph $G^2 = (V, E^2)$ such that $(u, v) \in E^2$ if and only if G contains a path with at most two edges between u and v .

Describe efficient algorithms for computing G^2 from G for both the adjacency-list and adjacency-matrix representations of G . Analyze the running times of your algorithms.

Problems Session 8

Graph traversal

Problem 1

Give a counterexample to the conjecture that if a directed graph G contains a path from u to v , then any depth-first search must result in $v.d \leq u.f$.

Problem 2

Give a counterexample to the conjecture that if a directed graph G contains a path from u to v , and if $u.d < v.d$ in a depth-first search of G , then v is a descendant of u in the depth-first forest produced.

Problem 3

Show that we can use a depth-first search of an undirected graph G to identify the connected components of G , and that the depth-first forest contains as many trees as G has connected components.

Data Structures and Algorithms

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Problems Session 9

BFS, Topological Sorting, MSP

Problem 1

Given a graph, traverse the graph using Breadth First Search and find the order in which nodes are visited.

Input: $\text{adj}[] [] = [[1, 2], [0, 2], [0, 1, 3, 4], [2], [2]]$

Problem 2

Do topological sorting for the following graph.

Input: $V = 6$, $\text{edges} = [[2, 3], [3, 1], [4, 0], [4, 1], [5, 0], [5, 2]]$

Problem 3

There are n cities and there are roads in between some of the cities. Somehow all the roads are damaged simultaneously. We have to repair the roads to connect the cities again. There is a fixed cost to repair a particular road.

Input is in the form of edges $\{u, v, w\}$ where u and v are city indices, w is the cost to rebuild the road between u and v . Print out the minimum cost to connect all the cities by repairing roads.

Input: $\{\{1, 2, 1\}, \{1, 3, 2\}, \{1, 4, 3\}, \{1, 5, 4\}, \{2, 3, 5\}, \{2, 5, 7\}, \{3, 4, 6\}\}$

Problems Session 10

Problems on Graphs

Problem 1

Check whether a given graph is Bipartite or not.

Input: $V = 4$, $\text{edges}[] = [[0, 1], [0, 2], [1, 2], [2, 3]]$

Problem 2

Detect a cycle in a directed graph.

Input: $\text{adj}[][] = [[1], [2], [0, 3], []]$

Problem 3

Number of Islands Given an grid consisting of 'L' (land) and 'W' (water), we need to count the total number of **islands** present in the grid without modifying the original grid. An island is defined as a group of connected 'L' cells that are adjacent **horizontally, vertically, or diagonally**, and surrounded by water or the boundary of the grid.

Examples:

Input:

$$\text{grid}[][] = \begin{bmatrix} 'L' & 'L' & 'W' & 'W' & 'W' \\ 'W' & 'L' & 'W' & 'W' & 'L' \\ 'L' & 'W' & 'W' & 'L' & 'L' \\ 'W' & 'W' & 'W' & 'W' & 'W' \\ 'L' & 'W' & 'L' & 'L' & 'W' \end{bmatrix}$$

Problems Session 11

Strassen's Algorithm

Problem 1

How quickly can you multiply a $kn \times n$ matrix and a $n \times kn$ matrix using Strassen's algorithm as a subroutine? Answer the same question with the order of the matrices reversed.

Problem 2

Show how to multiply the complex numbers $a + ib$ and $c + id$ using only three multiplications of real numbers. The algorithm should take a, b, c and d as input and produce the real component $ac - bd$ and the imaginary component $ad + bc$.